

Total No. of Questions : 8]

SEAT No. :

PC2462

[Total No. of Pages : 2

[6354]-590

B.E. (Information Technology)

AUGMENTED AND VIRTUAL REALITY

(2019 Pattern) (Semester - VIII) (Elective - VI) (414452B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain the significance of visual perception in virtual reality systems. **[9]**
b) Describe visual rendering process in detail. **[9]**

OR

- Q2)** a) Describe different techniques of motion in both the real and virtual worlds. **[9]**
b) What is tracking? Explain the Tracking in detail. **[9]**

- Q3)** a) How does Augmented Reality Work? Explain with a suitable example. **[9]**
b) Explain Ingredients of an Augmented Reality Experience. **[8]**

OR

- Q4)** a) What Does Augmented Reality Mean? Describe the augmented reality's history. **[8]**
b) Describe any two applications of Augmented Reality. **[9]**

- Q5)** a) Describe Sensor tracking methods in detail. **[9]**
b) Identify and discuss the various displays that are used in augmented reality systems. **[9]**

OR

P.T.O.

- Q6)** a) List different types of AR devices and explain the feature and uses of it. [9]
b) Describe sensor fusion with example. [9]

- Q7)** a) What is natural-feature tracking? Explain how it differ with marker-based Tracking. [9]
b) What is simultaneous localization and mapping in AR? [8]

OR

- Q8)** a) List and Explain the major augmented reality software components? [9]
b) What is marker tracking in AR? Explain in detail. [8]

